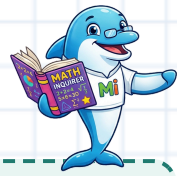
PICTURE EQUATIONS!

Name _____ Date _____

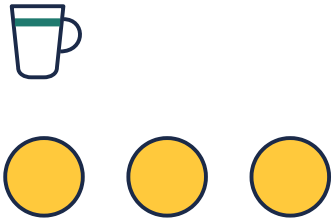


After question 4 you may choose: carry on, or try **Think harder**.

Each shape is a number. Use counters. Check your answer in the line.

No.S1

A cup hides some counters. 3 counters are outside the cup. There are 11 counters in all. Fill in the blank.



The cup holds ☐ counters.

No.S2

Each shape is a number. Find the number for the circle.

● + ■ = 10

■ = 4 ● = ☐

No.S3

Each shape is a number. Find the number for the heart.

♥ + ♥ = 16

♥ = ☐

No.S4

Each shape is a number. Find the number for the star.

★ + ★ + ★ = 15

★ = ☐

Name _____ Date _____

The same shape is always the same number. In our puzzles, different shapes are different numbers.

No.S5

Each shape is a number. The square is 4. Fill in the blanks.



(a) The 2 triangles make ☐☐.

(b) Each triangle is ☐.

No.S6

Ben says the circle is 7. Look at the picture.

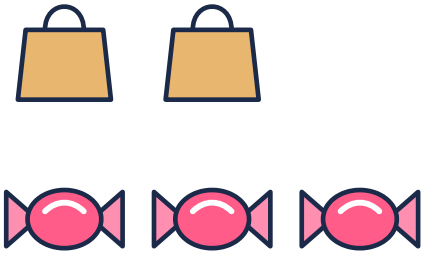


(a) Is Ben correct? Circle Yes or No. **Yes** **No**

(b) The circle is ☐.

No.S7

Each bag has the same number of sweets. 3 sweets are outside the bags. There are 13 sweets in all. Fill in the blanks.



(a) The 2 bags have ☐☐ sweets in all.

(b) Each bag has ☐ sweets.

REASONING ABOUT EQUALITY: PICTURE EQUATIONS

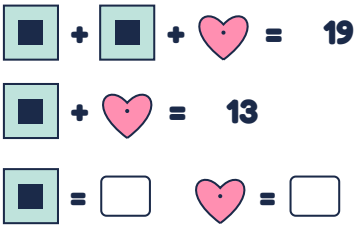
Think harder

Name _____ Date _____

Use counters or draw. Check your answer in every line. × means groups of. Choose any question.

No.T1

Each shape is a number. Find the number for each shape. Show your working.



No.T2

Kumar says the heart is 8 and the star is 7. Look at the picture. Decide if each statement is True or False.

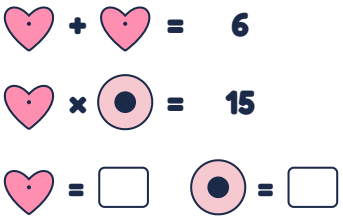


Statement	True	False
Line 1 is true with Kumar's numbers.	☐	☐
Line 2 is true with Kumar's numbers.	☐	☐

Put a tick (✓) in the correct box.

No.T3

Each shape is a number. Find the number for each shape.

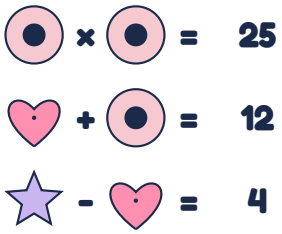


Name _____ Date _____

× means groups of. Choose any question. You do not have to finish them all.

No.G1

Each shape is a number. Find the number for each shape.



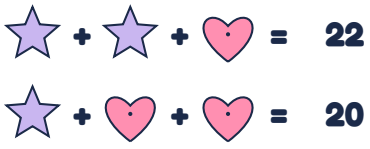
(a) The circle is ☐.

(b) The heart is ☐.

(c) The star is ☐.

No.G2

Each shape is a number. Fill in the blanks.



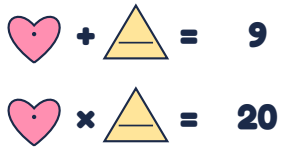
(a) One star and one heart make ☐☐.

(b) The star is ☐.

(c) The heart is ☐.

No.G3

Each shape is a number. The heart is greater than the triangle. Find the number for each shape.



(a) The heart is ☐.

(b) The triangle is ☐.

Name _____ Date _____

Investigation: One line or two?

First: The circle + the triangle = 10. One pair: circle 2, triangle 8. Write 2 more.

Next: The circle + the circle + the triangle = 14. Do any pairs fit both lines? Why?

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

Each shape is a number. The triangle + the triangle = 18. Find the number for the triangle.

The triangle is ☐.

No.E2

The heart + the square = 13. The square is 5. Find the number for the heart. Choose the correct answer. Write its number in the bracket.

(1) 5

(2) 6

(3) 7

(4) 8

( )

No.E3

Each shape is a number. The star is 5. Find the number for each circle.



Each circle is ☐.

Look-back

There are no right answers. You can pass. Nobody reads it out.

Something I did well: .....

The hardest part was ... ☐ finding the number for one shape ☐ using two lines ☐ checking in every line ☐ something else

What helped me: ☐ counters and shape cards ☐ drawing it ☐ checking my answer ☐ I was not stuck

Next time I will try: ☐ using counters ☐ checking my answer ☐ drawing it ☐ asking a question ☐ I am still stuck (my grown-up will photograph this page for my tutor)

☐ I would like to pass

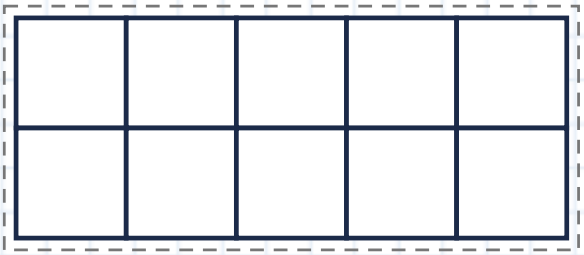
Name _____ Date _____

Kit page: ten-frames and counters

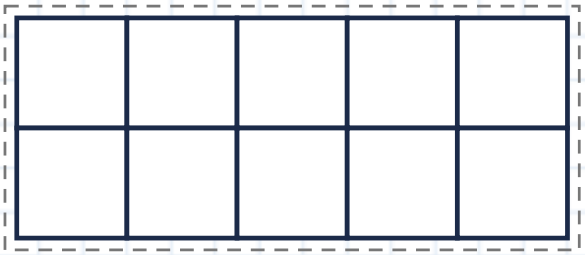
A grown-up cuts out the 20 counters (10 red, 10 yellow) before the lesson. Red counters can hide in a cup; yellow counters can be the loose ones. The ten-frames stay on the page.

Print this page on its own sheet, or on one side only, so cutting never spoils another page. A grown-up cuts; children do not use scissors. Keep the pieces in an envelope with your name. We use them again.

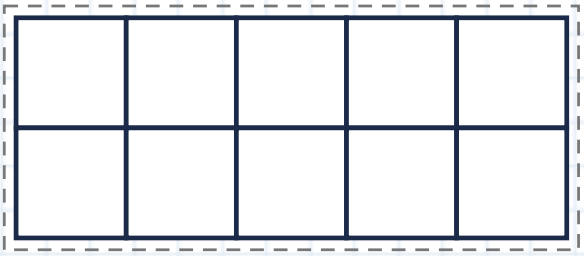
Ten-frame 1



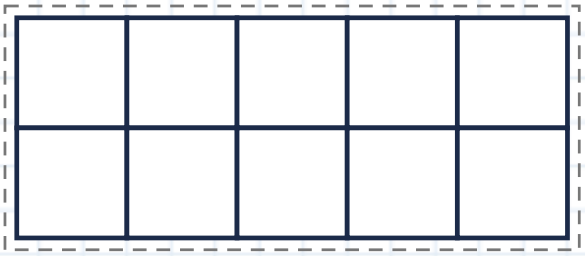
Ten-frame 2



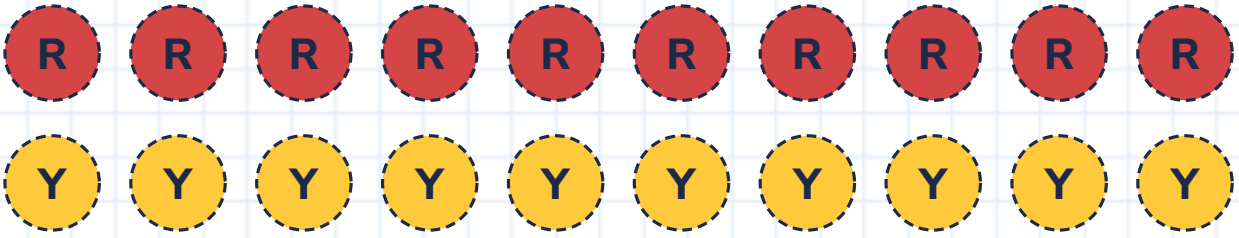
Ten-frame 3



Ten-frame 4



Counters: cut out all 20 (10 red marked R, 10 yellow marked Y)



Name _____ Date _____

Kit page: shape cards

A grown-up cuts out the 15 shapes before the lesson: three each of circle, square, triangle, heart and star. Each shape is drawn, so a child matches pictures, not words. A shape is big enough to hold 3 counters. Keep the shapes in an envelope.

Print this page on its own sheet, or on one side only, so cutting never spoils another page. A grown-up cuts; children do not use scissors. Keep the pieces in an envelope with your name. We use them again.

